Element A: Presentation and Justification of the Problem

Defining the Problem:
The problem is supposed to create a recyclable and effective book end that is able to sustain the weight of the books, not move and supports the Green School initiative.

Traditional bookends are typically made of plastic or metal, therefore are not recyclable, environmentally friendly and sustainable. The issues our stakeholder has with them is they don't last long as they rust easily, the books aren't always supported with the bookends, the lack of support within the bookends result in the books falling off and becoming slightly damaged. A reduced quality in the visual appeal of books will directly affect the stakeholder in which people will be less likely to borrow such books. Our stakeholder's bookends have uneven pressure on the book covers as it doesn't lie flat against the book, leading to wear and tear overtime. Along with keeping them looking simple yet elegant, it’s required we use recyclable materials to make them to follow our school’s Green School initiative.

These are traditional bookends that this project aims to improve upon. Our portfolio will go more into the strengths and weaknesses of each to analyze and decide upon our final design.

Problem Elaboration:
The problem occurs in the school library and impacts the organization, appealingness and preservation of the books.

- Books frequently fall or become misaligned, creating improper lines if they are not stored properly
- Solving this problem can help other stakeholders who use the library often like other students, staff and teachers like Mr.Eisenburg.

Identifying Stakeholders & Their Concerns:

The primary stakeholder is Ms.Tatro, our librarian who ensures that the books are stored properly and remain in good condition. She is the person who presented the problem to us, ensuring that books are stored properly and remain in good condition.

In addition, the people who use the library like students and faculty are our next primary stakeholders; we will talk to faculty like Mr.Eisenburg and students who consistently use the library.

These are some of the concerns of each of the primary stakeholders.

The School Media Specialist

- Longevity of the bookends
- Sustainability and support of the Green School Initiative
- Aesthetically Pleasing

Students

- Aesthetically Pleasing
- Able to sustain the weight of the books
- Easily moveable

Staff/Teachers

- Green School Initiative support
- Lightweight and supporting
- Sustainability

We also have several secondary stakeholders including the principal of our school, Dr.Barnes, who is responsible for upholding the integrity and appearance of the entire school. The functioning of the library indirectly affects the job of Dr.Barnes and therefore she is also a secondary stakeholder. In addition, the administrators at our school are also stakeholders because they regularly use the library and require upkeep.

<u>Design Requirements:</u>

- Keep books aligned and functional while remaining aesthetic and nice to look at.
- Be recyclable, lightweight and still yet effective to support the Green School initiative while accomplishing the final goal.
- Be lightweight yet sturdy, and evenly distribute pressure against the book covers to prevent damage
- Will not easily break, be damaged or rust: fixing the current problems Ms.Tatro has.

- Have a visually appealing aesthetic

In order to ensure our final design effectively addresses the problem, we will consult all of our stakeholders to guide our design process and refine our prototype based on real feedback.

Problem Justification:

The Green School Initiative are schools that "share a commitment to environmental sustainability" (National Association of Independent Schools "What Is a Green School?") pushes recycling and helping the planet. Its importance comes from the lack of recycling in the U.S. Lower than major economies, "America's national recycling rate is just 32%" (New York Times "I'm Appalled by What I Learned About Recycling. But We Can Fix It"). Recycling will "create a healthier planet for ourselves and future generations." (U.S. Environmental Protection Agency "Recycling Basics and Benefits"). Recycling is the only way to make sure the Earth continues to be habitable. Something such as using recyclable bookholders in a school library may seem too small, but small acts are what lead to big results. Making sure the school supports the Green School Initiative and uses green materials such as bookholders that work well by not moving and is able to sustain the weight of books is important both locally and globally.